

Computer Network Models Introduction:

- For data communication to take place and two or more users can transmit data from one to other, a systematic approach is required.
- This approach enables users to communicate and transmit data through efficient and ordered path.
- It is implemented using models in computer networks and are known as computer network models.

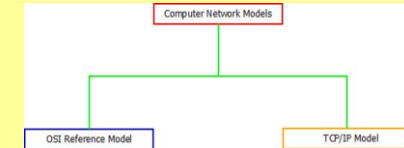
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Computer Network Models Introduction: (Cont:)

- Computer network models are responsible for establishing a connection among the sender and receiver and transmitting the data in a smooth manner respectively.

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Computer Network Models



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What is the OSI model ?

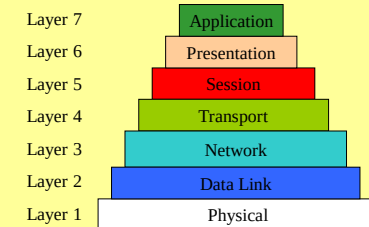
- Open Systems Interconnection model is fundamental to all communications between network devices.
- Developed in 1974 by *ISO* after the American Department of Defence began using the TCP/IP suite of protocols.
- Finally adopted in 1977. It is now the *theoretical model* for how communication takes place between network devices.

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- Can break communications functions into layers.
- Allows vendors to develop software and hardware to execute functions within each layer separately.
- Most common model is the Open Systems Interconnection (OSI) seven layer model.

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What are the seven layers ?



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Remembering the Layers

Application	All
Presentation	People
Session	Seem
Transport	To
Network	Need
Data Link	Data
Physical	Processing

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Memory Aid - Please Do Not Throw Sausage Pizza Away

Application	7	Away
Presentation	6	Pizza
Session	5	Sausage
Transport	4	Throw
Network	3	Not
Datalink	2	Do
Physical	1	Please

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Layers

- In the sense of purpose and responsibility, each layer is separate and independent
- Each has its own function, but also provides a service to those layers above and below itself
- The model should be considered an aid to understanding the nature of communication on the network - and useful in sorting out troubles that might occur on a network
- By providing, it allows both software engineers and hardware manufacturers ensure their products work together.

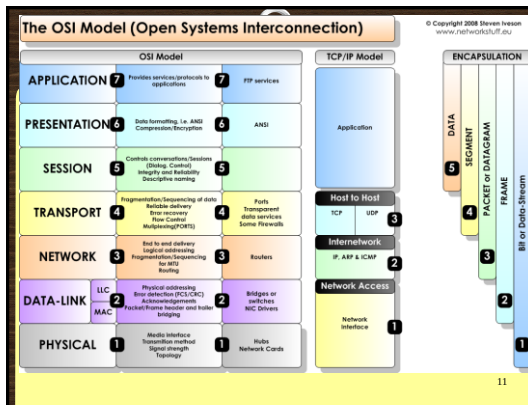
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The Layers at Work

As the next slide shows:

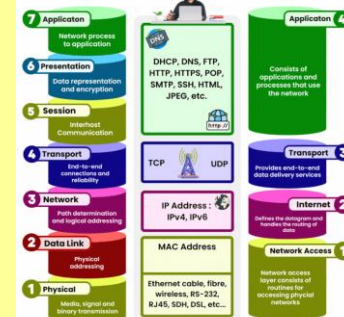
- When communicating, each OSI layer talks with the same layer in the other device
- E.g. the Application Layer of Device A communicates with the Application Layer of Device B, by passing the data through the other layers
- The Application Layer of each device is not concerned with how the other layers are functioning, but it does rely on them to do their job

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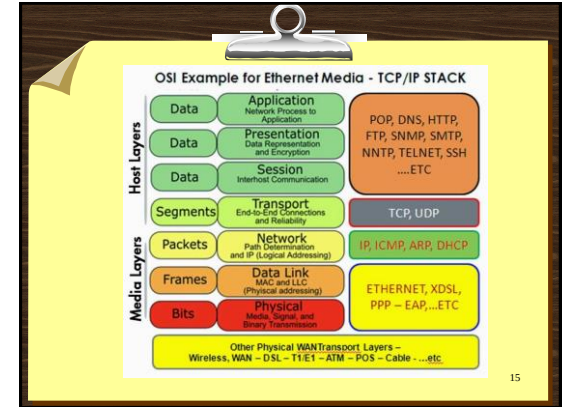
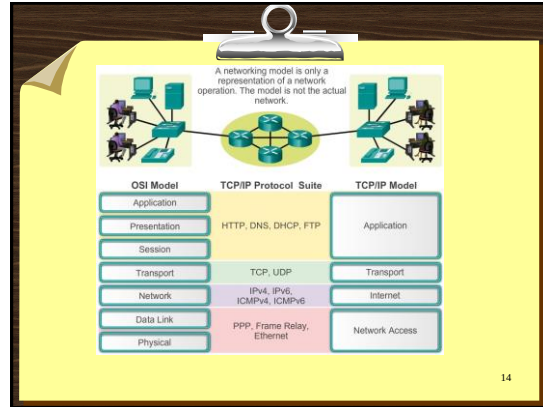
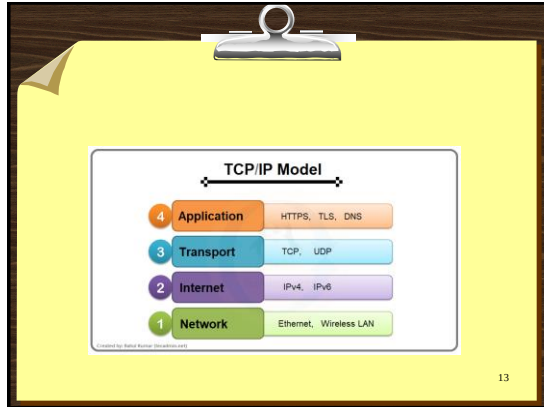


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OSI Model vs TCP/IP Model

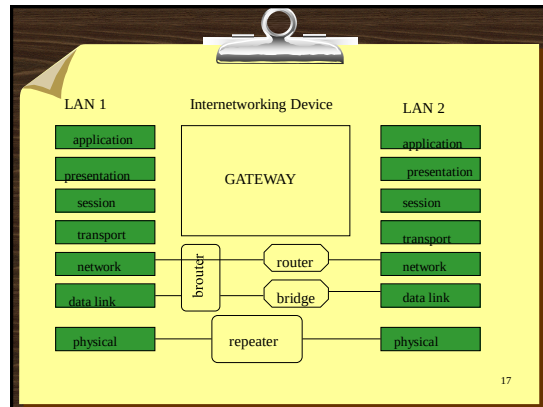


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Internet Protocols versus OSI 7-Layer Model

Layer	OSI	Internet	Data Format	Protocols
7	Application	Application	Messages or Streams	TELNET SNMP FTP
6	Presentation			
5	Session			
4	Transport	Transport or Host-Host	Transport Protocol Packets	TCP UDP
3	Network	Internet	IP Datagrams	IP
2	Data Link	Network	Frames	
1	Physical			



Network Models Summary

OSI Model	Internet Model	Groups of Layers
7. Application layer	5. Application layer	Application Layer
6. Presentation Layer		
5. Session Layer		
4. Transport Layer	4. Transport Layer	Internet Layer
3. Network Layer	3. Network Layer	Hardware Layer
2. Data link Layer	2. Data link Layer	
1. Physical Layer	1. Physical Layer	

How does data flow ?

When data is sent from the application on the source computer the following happens

- Data in the form of a packet moves down through the layers
- When it reaches the Physical Layer it is ready to be sent along the cable
- At the Physical Layer the bits may be analogue or digital, in the form of electrical, light or radio waves

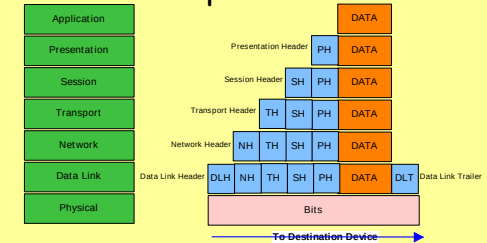
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How does data flow ? (2)

- The data is transmitted to the destination device
- It travels up through the layers of the OSI model, reaching the user.
- As data moves down through the layers it is encapsulated - ie additional information is added as headers or trailers
- The data in the packet does **not** change
- See the following diagram

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Encapsulation



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Packet - E-mail Example

Header	Sender's IP address Receiver's IP address Protocol Packet number	96 bits
Payload	Data	896 bits
Trailer	Data to show end of packet Error correction	32 bits

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Application Layer

- 'Closest' layer to the user
- Works with the applications you use to communicate over the network
- E.g.. Services include SMTP, HTTP and FTP
- Clicking on a link on a web page issues a command for the browser to retrieve the relevant information from the Internet
- In this example your computer is the source, and the host of the web site information is the destination
- The application completes your request and delivers the information to your computer

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Application Services

- File Services
- Electronic-mail Services
- Network-printing Services
- Application Services
- Database Services

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Presentation Layer

This layer has three fundamental functions

(1) Data Presentation

- Enables receiving device to understand the information sent from the source
- Converts data from native format (abstract syntax) to a common format (transfer syntax), e.g. ASCII

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Presentation Layer (2)

(2) Data Compression

- By reducing the volume of data, transfers can take place in less time
- Packets are examined and such things as spaces in text removed
- The destination device returns the data to its original format before passing to the Application Layer

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Presentation Layer (3)

(3) Data Encryption

- Allows data to be converted to a form which hides its meaning, apart from those you wish to see it
- Not all data is encrypted on its journey across the network
- In order for decryption to occur at the destination device a 'key' is required

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Session Layer

- Primarily responsible for handling the session between devices (beginning, maintaining and finishing)
- Enforces order in the communication between devices
- Regulates the flow of data
- It takes responsibility for the following

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Session Services

The following services are provided:

- Establishing a Connection
- Maintaining the Session
- Ending the Connection
- Dialogue Control
- Dialogue Separation

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Session Services (2)

- Handshaking - SYN and ACK packets
- 'Keep alive messages'
- Session must be terminated (otherwise one device will be still transmitting without any device actually listening)
- Dialogue Control (simplex, half-duplex, full-duplex)
- Dialogue Separation - checkpoints within the transmission which allow the detection of lost packets, and subsequent re-transmission

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Transport Layer

- Ensures reliable transport of packets from source to destination
- Also manages the speed of transmission - flow control
- There are two types of transmission (Connection-Oriented Transmissions and Connectionless Transmissions) - see next slide

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Connection-Oriented Transmissions

- Also known as 'Reliable Transport Method' - uses acknowledgement (ack) packets on successful receipt of data
- Extra packets slows down communication
- Features are
 - Reliability
 - Slower Communication
 - Packets are re-transmitted if unrecognisable or not received
- Once all the data is received successfully, the packet is re-assembled and the Transport Layer passes it to the Session Layer

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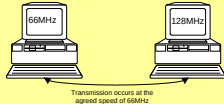
Connectionless Transmissions

- In this mode the transmitting device does not require acknowledgements from the receiver, and continues to transmit on the assumption that the data was received
- Features are:
 - Little or No Reliability
 - Faster Transmission
 - Packets are not Re-transmitted

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Flow Control

- Establishes the maximum speed at which both sender and receiver can communicate at
- Transport Layer determines largest packet size which can be sent
- Packets are numbered - to allow re-assembly in the correct order



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Network Layer

- Responsible for the correct addressing and delivery of packets of data
- These are known as datagrams
- Uses the network address (this is a logical address - and does not depend upon any hardware in the device, or the device's physical location)

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Network Layer (2)

The Network Layer does the following:

- Adds the address to the packet (encapsulation)
- Maps the network address to the device's physical address
- Determines the best path for the packet (routing)
- Ensures that the packet is in the correct format for the destination

See Diagram =>

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How does it work ?

- Encapsulation at the Transport Layer involves adding the address of the sender to the datagram
- The destination address is now added. Both addresses are **logical**.
- Both addresses are necessary for packets to move between end systems.
- If a packet must move to another network, a routing protocol is required
- If different packet lengths are used on the different networks, the Network Layer formats the data accordingly
- The primary piece of hardware which works on this layer is the router. (covered in detail, later in course)

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Data Link Layer

- Has two sub layers of its own:
 - Logical Link Control (LLC)
 - Media Access Control (MAC)
- LLC acts between protocols such as Internet Protocol (IP) and the MAC method.
- MAC is responsible for the connection to the physical media (eg cable)

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MAC

- Each NIC has a unique number hard coded in to the card - its physical address
- The first 6 digits denote the manufacturer, the next six are unique) - type "winipcfg" on your PC
- When the MAC address is added to the packet it is now known as a frame
- It now has all the information required to travel from the source to the destination

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Physical Layer

- The lowest, bottom, layer - responsible for the physical connection between devices
- The NIC converts the data (bits) in to transmission signals.
- Transmissions may be analogue or digital
- Responsible for the rate of transmission
- Includes all components such as the type of connector (RJ-45, Token Ring, BNC, SC connector)
- Devices at this level include NICs, repeaters, hubs and concentrators

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Computer Network Models : Advantages Of OSI Model

- The OSI Model being one of the most widely used computer network models does possess some major advantages which makes it so popular. These are :
- Each layer has its definite structure and functionality which makes OSI model simple and easy to use.
- It is a general purpose reference model that can be used for data communication.
- Connection oriented and connection-less services are supported.
- Connection between any type of devices or host or hardware or software is possible.

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Computer Network Models Disadvantages Of OSI Model

- Because of its inability to fit protocols, this model was replaced by TCP/IP Internet Model.
- Session and Presentation layers does not provide high end functionalities and are not of much use as compared to other layers.
- Connection oriented and connection-less services are supported.
- Connection between any type of devices or host or hardware or software is possible.

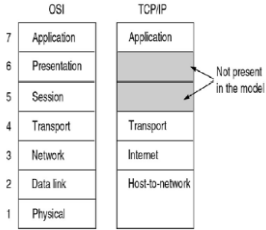
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1. Physical Layer	1. Physical Layer	Layer

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Reference Models (2)



The TCP/IP reference model

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